

Pratik K

AI & Machine Learning Engineer | Full-Stack | Algorithmic Trading

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Education

Bachelor of Technology, Computer Science and Engineering
International Institute of Information Technology, Hyderabad, India

May '16 - May '21

Experience

Full-Stack / AI Engineer *MediBank (Remote)*

Jan '26 - Feb '26

- Built healthcare document portal (React + FastAPI) with doctor/patient auth, patient search, and MID-based report access
- Implemented AI document pipeline (Vertex AI) that classifies uploads and runs structured extraction for lab reports; designed status flow and versioning in MongoDB
- Delivered report viewing and lab insights UI: unified dialog, PDF/image streaming, and dedicated lab-report view with metadata, panels, and abnormal value highlighting
- Set up Docker Compose stack (Nginx, React, FastAPI, PostgreSQL, MongoDB) with encrypted artifact storage and streaming

Full-Stack Engineer *TraderRoyale (Trade Wise Capital LLC) (Remote)*

Jul '25 - Feb '26

- Built real-time competitive simulated forex trading platform from scratch: backend API, market data pipeline, order execution engine, competitions, PvP, and React frontend
- Implemented real-time market data pipeline with Polygon.io, Redis Streams, TimescaleDB, and WebSocket broadcast; added leader election for multi-worker deployment
- Designed order execution engine: tick-driven market/limit/stop orders, multi-level SL/TP with OCO, partial fills, and real-time event broadcast
- Delivered competition and PvP systems with crypto payments (USDT ERC-20), MetaMask checkout, tx verification, and claim/payout flows

Full-Stack Developer *PostAI (Trade Wise Capital LLC) (Remote)*

Aug '25 - Nov '25

- Built PostAI, AI-oriented social media management product enabling scheduled publishing to Facebook and Instagram via OAuth, media uploads, and asset library
- Implemented Facebook and Instagram OAuth and Meta Graph API integration for account linking, token refresh, and multi-account support with mobile-friendly flows
- Delivered resumable media uploads, URL-based media for Instagram, and platform-specific validation; refactored post scheduling APIs and multi-post creation from a single dialog
- Shipped full-stack app with React (Vite) and FastAPI; PostgreSQL and Redis with RQ for scheduled jobs; landing page, ToS, Privacy Policy, and mobile-responsive UX

AI Innovation Specialist *Trilogy (ESW Capital Group) (TX, USA)*

Feb '25 - Present

- Leading high-impact AI-first projects, applying advanced machine learning and deep learning techniques to solve complex enterprise software challenges
- Driving innovation within the software portfolio using PyTorch, TensorFlow, and modern ML frameworks

Quantitative Engineer *Trade Wise Capital (USA)*

Nov '22 - Present

- Architected and built complete end-to-end algorithmic trading bot as scalable VPS SaaS solution using Python, achieving consistent alpha generation
- Developed machine learning models for supplemental trading signal generation, improving strategy performance by 15%
- Research and integrate new alpha-generating strategies using ML-driven indicators on MT5 and TradingView platforms

Partner, CTO *Synalytica LLC (USA)*

Nov '20 - Present

- Co-founded venture studio, leading technology vision for AI-driven products across healthcare, finance, and education sectors
- Architected FolioWiz: AI-powered stock portfolio optimization platform using machine learning for risk-adjusted returns
- Built Starter Up: AI-based content creation tool leveraging NLP and generative models
- Developed Energylicious: predictive analytics platform for energy consumption optimization using time-series forecasting
- Created WarrVault: logistics tracking system with computer vision for warranty document processing

Chief Technical Lead *QuantifAI (Malaysia)*

Apr '19 - Feb '20

- Engineered high-frequency, low-latency algorithmic trading platform using FiX API, handling 1000+ transactions per second

- Deployed autonomous forex trading strategies on AlgoTrader platform, achieving 23% annualized returns through ML-driven backtesting
- Led and mentored team of 5 software engineers, establishing CI/CD pipelines and best practices for financial ML systems

Technical Consultant/Contractor *Docturnal (Hyderabad, India)*

May '18 - Feb '20

- Re-architected platform transitioning to Airflow-based ML pipeline and scheduling system, reducing processing time by 40%
- Provided architectural consultation for Python microservices platform, implementing FastAPI and Docker containerization
- Designed ML architecture using XGBoost and CNNs for respiratory illness classification (COPD, tuberculosis) from cough audio, achieving 87% accuracy

Research Experience

Research Scholar *Cognitive Research Lab, IIIT-H (India)*

Dec '18 - May '21

Developed theoretical framework for modeling real-world social networks using graph neural networks and network science. Research focused on mental illness detection through machine learning analysis of music consumption patterns and social interaction trends. Published findings on correlation between social media behavior and mental health indicators, achieving 82% classification accuracy using ensemble methods.

Deep Learning Intern *Siemens, Perception Lab (Bangalore, India)*

May '19 - Jun '19

Contributed to 'ITF' intelligent traffic management system deployment. Optimized performance-critical computer vision pipeline by converting Python implementation to C++ using OpenCV and TensorRT, achieving 3x speedup. Containerized application using Docker for deployment on Nvidia Jetson TX2 edge devices, enabling real-time traffic analysis at 30 FPS with <100ms latency.

Deep Learning Intern *Valeo Dar (Germany)*

Jul '18 - Jan '19

Advanced computer vision research for autonomous driving systems. Developed fisheye stereo correspondence algorithms and direct depth estimation models using CNNs to replace expensive LIDAR sensors. Engineered production-grade C++ deep learning pipeline achieving 95% depth estimation accuracy at 20 FPS, reducing sensor costs by \$10K per vehicle while maintaining SLAM module performance.

Vision Researcher *Computer Vision Lab, IIIT-H (India)*

May '18 - Dec '18

Conducted research on mobile camera image quality assessment under Dr. Vineet Gandhi. Developed novel algorithms for computational photography: bokeh detection using semantic segmentation and depth estimation, plus quantitative bokeh quality metrics. Implemented models using PyTorch achieving 89% accuracy on smartphone image dataset of 10K+ photos.

Machine Learning Engineer *Variance AI*

Sep '17 - Mar '18

Architected core ML framework for AI-powered fitness training platform. Implemented real-time 3D pose estimation and human pose tracking using Kinect depth sensors and point cloud processing. Developed user recognition system and automated exercise correction feedback using sequence-to-sequence models, achieving 91% accuracy in pose classification across 50+ exercise types.

Projects

Pill Anomaly Rejection System *Graviti Pharma*

Jul '20 - Dec '20

- Architected end-to-end computer vision pipeline using CNNs and PyTorch for real-time anomaly detection. Built custom dataset of 50K+ pill images with defect annotations.
- Achieved 96% accuracy in defect classification, processing 200 pills/second with sub-50ms latency, reducing manual inspection costs by 65%.

Tangible Learning *Stanford University*

Oct '18 - Dec '19

- Developed mixed-reality application using computer vision and deep learning to recognize physical learning tiles and project them in real-time to remote participants. Implemented custom object detection model using TensorFlow and integrated with WebRTC for low-latency streaming.
- Enabled seamless tangible interaction for distributed teams with <100ms end-to-end latency, supporting 10+ concurrent users.

Hindi-English Code Mixed Dataset *Language Tech Research Center, IIIT-H*

Sep '18

- Built high-performance multi-processed web scraping pipeline using Python and BeautifulSoup to collect and process code-mixed social media content. Implemented distributed processing across 16 cores with Redis queue management.
- Collected dataset of 7000+ articles (5M+ tokens), published for academic research, enabling sentiment analysis and language identification models for code-mixed text.

Competitive Coding Application *Project under Dr. Raghu Reddy*

Apr '17

- Developed full-stack web application using Flask with sandboxed Docker containers for safe code execution. Implemented custom judge module supporting Python, C++, and Java with automated test case generation and performance metrics.
- Platform used by 500+ students for competitive programming practice, executing 10K+ code submissions with 99.9% uptime.

Legend of Zelda Game Project *under Dr. Avinash Sharma*

Dec '17

- Developed 3D game remake using OpenGL and C++ with intelligent enemy NPCs featuring A* pathfinding, behavior trees, and state machines. Implemented custom shader programs for lighting and created procedural terrain generation.
- Fully playable game with 5 levels, showcasing advanced AI behaviors and achieving 60 FPS on mid-range hardware.

Linux Shell Project *under Dr. P.K. Reddy*

Sep '17

- Created feature-rich Zsh-like shell in C++ supporting process management, I/O redirection, piping, background jobs, and command auto-completion using trie data structure. Implemented signal handling and custom prompt themes.
- Fully functional shell with 25+ built-in commands, demonstrating deep understanding of OS concepts and system-level programming.

Skills

LLM : OpenAI (Text + Realtime (audio) + Agentic), Gemini (Realtime (audio + video) + text + agentic), Browser Use, LangChain, CrewAI

Machine Learning : PyTorch, OpenCV, Tensorflow 1.x, Vertex AI, PyTorch C++, Scikit-learn, XGBoost, Keras

Programming Languages : Python, Pinescript, MQL4/5, C++, C, Java, ECMAScript, TypeScript, Shell scripting, Matlab, PHP, Assembly (MIPS)

Web Development : FastAPI, React, Vite, Vue.js, Django, Flask, Laravel, Node.js, Express.js, Javascript, HTML5

Databases : PostgreSQL, MongoDB, Redis, TimescaleDB, MySQL, Neo4j, XQuery

DevOps & Cloud : Docker, Docker Compose, Nginx, Google Cloud Platform (GCP), Amazon Web Services (AWS ECS), Azure Cloud, CI/CD

Selected Coursework

Machine Learning I*
Artificial Intelligence
Game Theory*
Complexity and Advanced

Algorithms*
Algorithms
Data Structures
Structured Systems Design

Introduction to Databases
Operating Systems
Computer Organisation

* Denotes advanced or honors-level coursework

Other Experience

Teaching Assistant *IIIT Hyderabad*

Aug '18 - Nov '18

Served as TA for Structured Systems Analysis Design (SSAD) course. Guided three student teams in managing client requirements and software engineering best practices.

Technical Project Manager *MBR Informatics*

Jan '18 - Jul '18

Managed team of 10 developers, overseeing project timelines, CI integration, and workflow design. Led development of Django-based health management platform.

Web Administrator *IIIT Hyderabad*

Oct '17 - Jan '18

Managed entire college web framework and led design and implementation of new image-sharing portal for student body.

Web Developer *Mayabazaar*

Aug '16 - Oct '16

Developed user interface for movie industry-based e-commerce portal. Integrated secure payment gateway (Stripe) for online transactions.

Miscellaneous

- OSDG Volunteer: Led rebuild of college applications (courier portal, recommendation system) using modern technologies (2020)
- BeeTech Co-founder: Started Medium publication to provide consolidated free information about tech
- Teacher, Ashakiran (2017-2018): Taught English, Physics, and Science to underprivileged children
- Listener, 7 Cups (2016-2017): Provided emotional support through active listening sessions
- IIIT-H MUN: Member of the Organizing Committee (2016-2017)